

Analyzing and Visualizing Regional Sales Performance

Excel Mini Project Report

Submitted by: Sri Rajakavika Palanivel
Course: Data Verse Pro
Institute: Vinsup Skill Academy

Introduction

ABC Electronics operates across multiple regions and product categories. This project analyzes 4,000 sales records using Microsoft Excel to evaluate regional sales performance. The work included data cleaning, formula-based analysis, Pivot Tables, charts and an interactive dashboard that helps management monitor key metrics, compare regional performance and identify product category trends for better business decisions.

Objectives

- To clean and organize the sales dataset.
- To standardize text values using Excel functions.
- To calculate important business metrics using formulas.
- To analyze sales by region and product category.
- To create Pivot Tables for summarizing data.
- To build an interactive dashboard using charts and slicers.
- To generate useful business insights from the dataset.

Dataset Description

The dataset contains Order ID, Order Date, Region, Product Category, Sales Amount, Quantity Sold, Discount (%) and Profit. The data consisted of 4,000 records in CSV format.

Dataset Columns

Column Name	Description
Order ID	Unique order number
Order Date	Date when the order was placed
Region	Sales region

Product Category	Product type
Sales Amount	Total sales value
Quantity Sold	Number of products sold
Discount (%)	Discount given
Profit	Profit earned

Data Cleaning and Pre-processing

- The first step of the project was to clean the data.
- The following tasks were performed:

Removing Extra Spaces

The TRIM() function was used to remove unnecessary spaces from text values.

Standardizing Text

The UPPER() and LOWER() functions were used to make Region and Product Category names consistent throughout the dataset. And also SUBSTITUTE(), FIND AND REPLACE functions to correct the inconsistent data in region and product category columns.

Filtering Data

The dataset was filtered to find orders placed in the **South region** under the **Electronics** category during the last year.

Data Validation

The dataset was checked for blank values, formatting issues, and duplicate records to ensure accurate analysis.

After completing these steps, the dataset became clean and ready for analysis.

	A	B	C	D	E	F	G	H		A	B	C	D	E	F	G	H
1	Order ID	Order Date	Region	Product Category	Sales Amount	Quantity Sold	Discount (%)	Profit	1	Order ID	Order Date	Region	Product Category	Sales Amount	Quantity Sold	Discount (%)	Profit
2	ab899d2f	2023-04-06	norTh	groceries	859.29	44	37	1653.87	2	ab899d2f	2023-04-06	North	Groceries	859.29	44	37	1653.87
3	73484eb1	2024-04-30	MesSt	groceries	1546.81	23	33	1932.71	3	73484eb1	2024-04-30	West	Groceries	1546.81	23	33	1932.71
4	19395924	2024-03-12	eAsT	groceries	4703.58	31	15	527.47	4	19395924	2024-03-12	East	Groceries	4703.58	31	15	527.47
5	b3fde3d	2024-02-09	eAsT	groceries	3377.17	47	3	482.65	5	b3fde3d	2024-02-09	East	Groceries	3377.17	47	3	482.65
6	6e4bdd51	2024-07-29	\$ouTH	Cloth ing	3517.49	6	28	748.72	6	6e4bdd51	2024-07-29	South	Clothing	3517.49	6	28	748.72
7	21d88d69	2023-08-26	eAsT	FURNiture	362.55	19	50	314.71	7	21d88d69	2023-08-26	East	Furniture	362.55	19	50	314.71
8	9da2704d	2024-10-07	eAsT	sports	4811.17	22	18	1372.89	8	9da2704d	2024-10-07	East	Sports	4811.17	22	18	1372.89
9	a1bd6815	2023-03-15	MesSt	electronicS	3565.2	22	16	1437.3	9	a1bd6815	2023-03-15	West	Electronics	3565.2	22	16	1437.3
10	0bbd51fc	2023-06-29	eAsT	Cloth ing	3805.25	19	3	1537.28	10	0bbd51fc	2023-06-29	East	Clothing	3805.25	19	3	1537.28
11	8bf7e4b3	2023-03-13	\$ouTH	Cloth ing	2178.72	15	24	1696.65	11	8bf7e4b3	2023-03-13	South	Clothing	2178.72	15	24	1696.65
12	e6cf1255	2024-05-27	MesSt	groceries	2844.93	34	12	1166.39	12	e6cf1255	2024-05-27	West	Groceries	2844.93	34	12	1166.39
13	3ba1fc5c	2023-12-31	MesSt	electronicS	3600.04	10	2	1632.75	13	3ba1fc5c	2023-12-31	West	Electronics	3600.04	10	2	1632.75
14	388ff70d	2023-11-09	eAsT	\$portS	941.88	13	28	646.49	14	388ff70d	2023-11-09	East	Sports	941.88	13	28	646.49
15	ce2e9002	2022-12-24	\$ouTH	\$portS	3300.86	1	48	86.9	15	ce2e9002	2022-12-24	South	Sports	3300.86	1	48	86.9
16	a4f9a338	2024-02-07	eAsT	groceries	1081.62	4	19	611.06	16	a4f9a338	2024-02-07	East	Groceries	1081.62	4	19	611.06
17	eadceaa8	2024-01-27	eAsT	FURNiture	3455.67	39	21	208.1	17	eadceaa8	2024-01-27	East	Furniture	3455.67	39	21	208.1
18	ad7f9de9	2023-06-01	\$ouTH	groceries	2381.51	16	39	1083.96	18	ad7f9de9	2023-06-01	South	Groceries	2381.51	16	39	1083.96
19	35358e9c	2024-06-28	norTh	Cloth ing	2784.79	35	40	999.6	19	35358e9c	2024-06-28	North	Clothing	2784.79	35	40	999.6
20	31676338	2023-11-16	\$ouTH	Cloth ing	1082.98	8	11	1604.19	20	31676338	2023-11-16	South	Clothing	1082.98	8	11	1604.19
21	05d72e20	2024-09-21	eAsT	Cloth ing	787.79	47	18	1313.33	21	05d72e20	2024-09-21	East	Clothing	787.79	47	18	1313.33
22	68dbe328	2024-10-01	\$ouTH	FURNiture	312.7	36	43	559.52	22	68dbe328	2024-10-01	South	Furniture	312.7	36	43	559.52
23	78d3f47b	2023-05-31	eAsT	\$portS	265.32	21	36	595.91	23	78d3f47b	2023-05-31	East	Sports	265.32	21	36	595.91
24	9e4d1fe0	2024-04-18	MesSt	FURNiture	1253.64	35	14	1653.66	24	9e4d1fe0	2024-04-18	West	Furniture	1253.64	35	14	1653.66
25	9215a70e	2024-02-19	MesSt	Cloth ing	1874.07	28	23	1783.26	25	9215a70e	2024-02-19	West	Clothing	1874.07	28	23	1783.26
26	54ca3433	2023-04-25	eAsT	Cloth ing	2561.71	44	10	1007.97	26	54ca3433	2023-04-25	East	Clothing	2561.71	44	10	1007.97

Data Analysis

SUMIF, AVERAGEIFS, COUNTIF, FILTER and UNIQUE functions were used. Pivot Tables summarized sales and profit by region and category. Charts were created to visualize sales distribution and trends.

SUMIF()

Used to calculate the total sales for each region.

AVERAGEIFS()

Used to calculate the average discount percentage and average profit for the Furniture category.

FILTER()

Used to extract specific records based on conditions.

UNIQUE()

Used to identify unique values in the dataset.

COUNTIF()

Used for counting records that satisfy specific conditions.

These formulas helped summarize the sales data effectively.

Pivot Tables

Pivot Tables were created to summarize the dataset without changing the original data.

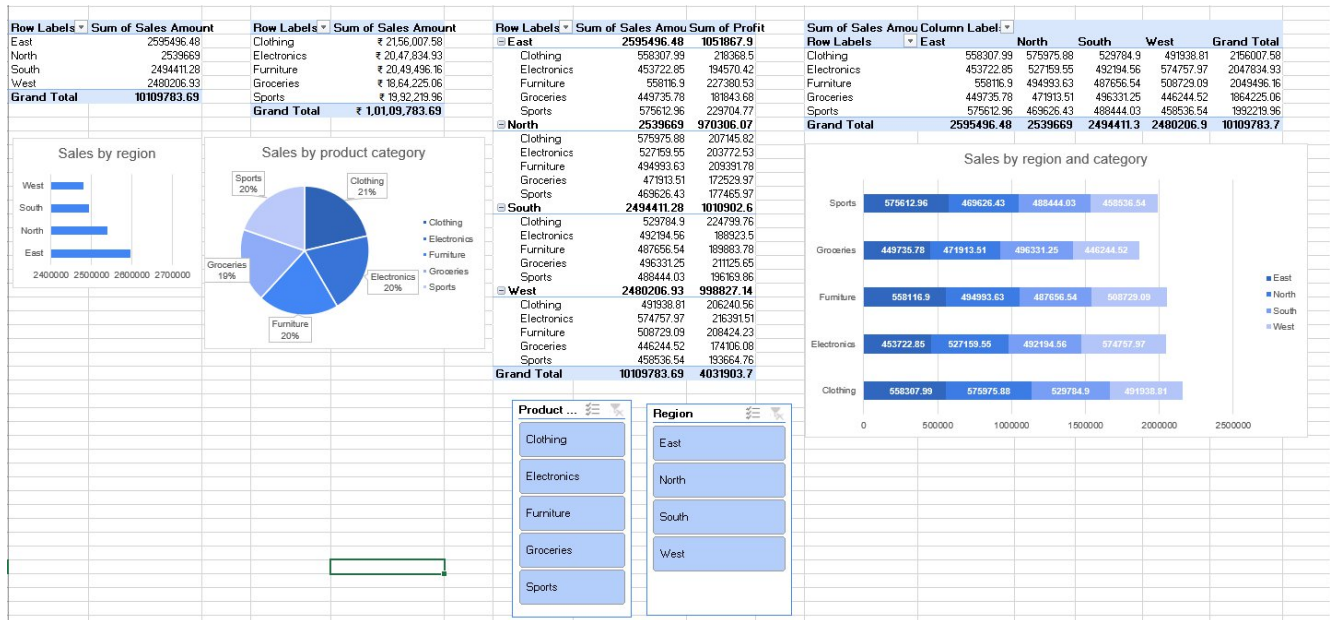
The following Pivot Tables were prepared:

- Sales by Region
- Profit by Region
- Sales by Product Category
- Sales by Region and Product Category

Slicers were added so that the dashboard could be filtered dynamically by Region and Product Category.

- The pivot tables were used to create charts based on summarization available in pivot tables.
- Each pivot chart represents different types of charts created.
- Slicers were created to change the values in all charts according to the data we select in slicer.
- This helps to analyze dynamically on each category and region.

The pivot tables and their respective charts are given below:



Dashboard

The dashboard includes KPI cards (Total Sales, Total Profit, Total Orders and Highest Selling Category), Sales by Region, Sales by Product Category, Region vs Category chart, Year-wise Sales Trend, Profit by Region and interactive slicers.

The final dashboard was designed using a pastel pink color theme with a clean layout.

The dashboard contains four KPI cards:

- Total Sales
- Total Profit
- Highest Selling Category
- Total Orders

It also includes different charts to represent the data visually.

Sales by Region

A horizontal bar chart compares sales among East, West, North, and South regions.

Sales by Product Category

A pie chart displays the percentage contribution of each product category to total sales.

Sales by Region and Product Category

A stacked bar chart compares sales distribution across different regions.

Year-wise Sales Trend

A line chart shows sales performance of different categories over multiple years.

Profit by Region

A doughnut chart displays the contribution of each region to the overall profit.

Region and Product Category slicers were added to make the dashboard interactive and user-friendly.

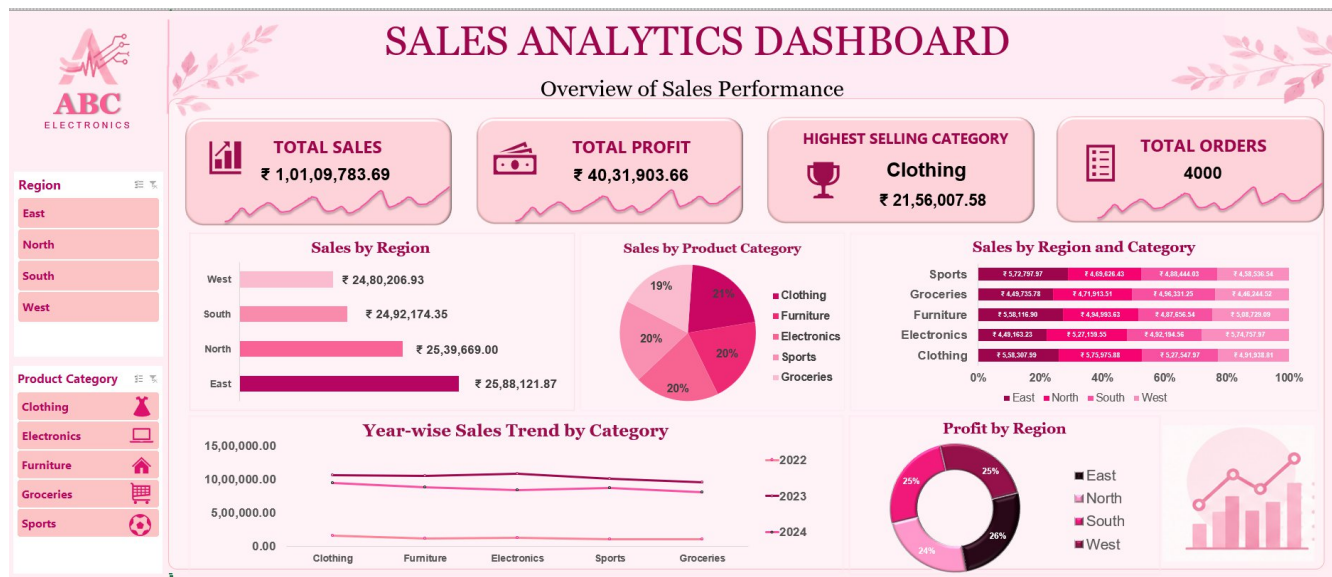


Figure 1: Interactive Sales Performance Dashboard

Business Insights

- The **East region** recorded the highest sales, indicating strong market performance.
- **Clothing** emerged as the highest-selling product category, contributing the maximum revenue.
- The company achieved a total sales of **₹1.01 Crore** and a total profit of **₹40.32 Lakhs** from **4,000 orders**.
- Sales and profit were distributed across all regions, showing a stable business presence.
- The interactive dashboard enables quick analysis of sales performance using Region and Product Category slicers, supporting better business decisions.

Recommendations

- Maintain adequate inventory for high-demand product categories like Clothing.
- Focus marketing efforts on lower-performing regions to improve sales.
- Use the dashboard regularly to monitor sales trends and support data-driven decision-making.

Challenges Faced

Cleaning inconsistent text, organizing Pivot Tables and designing a clean dashboard layout were the main challenges. These were resolved through careful data pre-processing and repeated testing.

Conclusion

This project improved my understanding of Excel-based data analysis and dashboard development. It demonstrated how raw data can be transformed into meaningful business insights for decision-making.